The Cosmic Microwave Background

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1. Starry Background: https://unsplash.com/photos/photography-of-starry-sky-uhjiu8FjnsQ
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12. Gravitational Potential Well: https://en.wikipedia.org/wiki/File:GravityPotential.jpg
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THE COSMIC MICROWAVE BACKGROUND

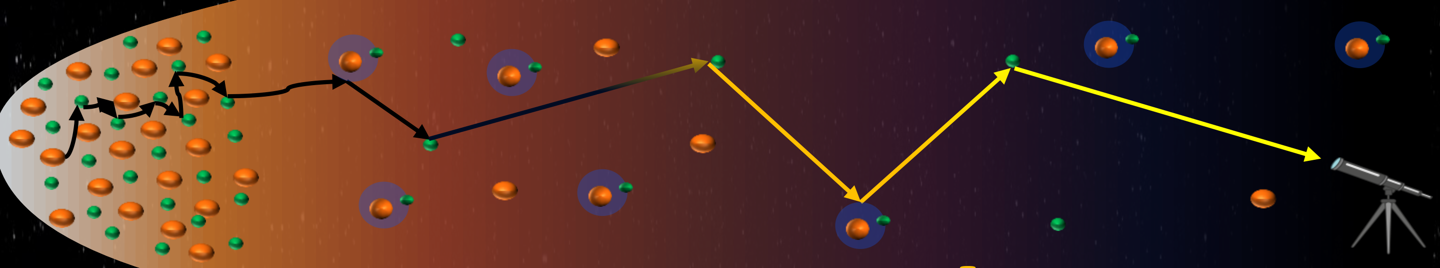
The Cosmic Microwave Background (CMB) is residual radiation from the creation event that permeates the universe uniformly in all directions. Looking into the distant universe is essentially peering into the past. As such, CMB represents a time 380,000 years after the Big Bang, and provides a snapshot of what the early universe looked like and how it evolved. Photons from this time have since ceased scattering off from matter and now reach us unhindered. The expansion of the universe has stretched the wavelengths of these photons into the microwave region of the electromagnetic spectrum.

1 The Early universe

An intensely hot and dense sea of ionised gas and electrons. The universe is opaque and unobservable since the photons are trapped and cannot escape to reach us.

3 Photon Decoupling Era

Electron density decreases. Frequency of photon-electron interaction decreases.



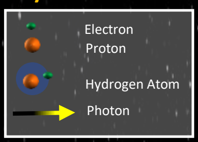
2 Recombination Era

The universe cools down enough to allow electrons and protons to combine and form light elements like hydrogen.

4 Last Scattering Era

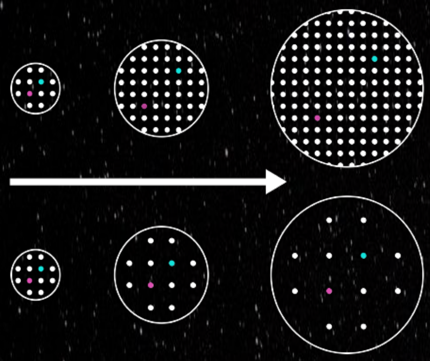
The last time a CMB photon interacts with an electron. Universal expansion becomes so fast that photons do not find an electron to interact with again. They are free to move and reach us.

Key



Steady-State Theory

Big Bang Theory



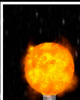
The **Steady-State** and the **Big Bang** are two competing theories concerning the origin of the universe. The Steady-State theory suggests an infinitely old universe that has experienced no change in its average density, with matter being created as the universe expands. The Big Bang theory proposes a denser universe closer to the origin, with matter getting diluted as the universe expands. The existence of CMB cannot be explained by the steady-state model as it implies that the universe has remained unchanged from the past to the present. Conversely, the CMB supports the Big Bang model, being the afterglow of the creation event.

Properties of CMB

The CMB is the most perfect blackbody* spectrum found in nature.

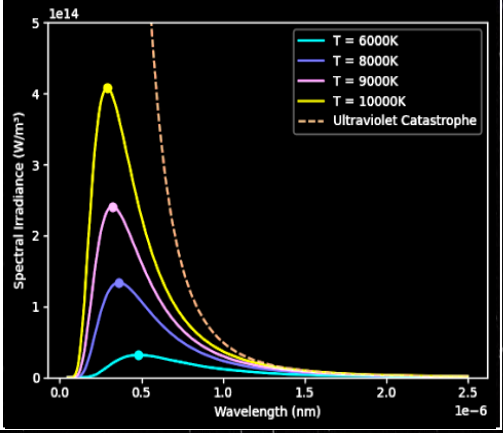
It has a temperature of ~2.725K which is nearly uniform throughout the universe regardless of the direction in which it is observed.

Despite this uniformity, small temperature fluctuations of ~20μK can be observed. These fluctuations are of paramount importance.



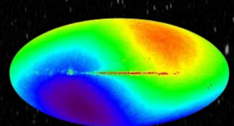
***Recall**

An ideal blackbody is a perfect emitter and absorber of radiation of all wavelengths.

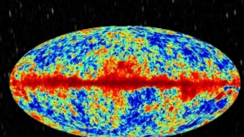


CMB Maps

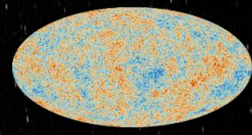
Hot Cool



The hot and cold dipole is due to the Earth's movement relative to the CMB.

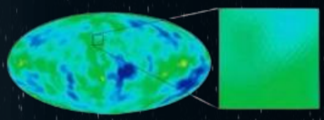


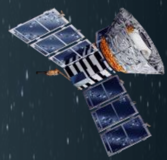
The dipole has been removed, showing a central red band. This is the microwave signal from the Milky Way.



The interference from the Milky Way is removed. This is the actual CMB map.

NASA COBE Mission 1989



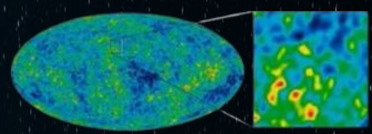


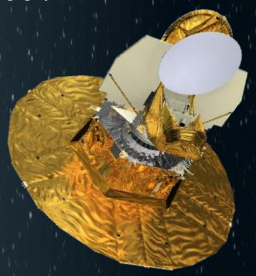
Discovered that CMB spectrum closely resembled the spectrum of an ideal blackbody and that its temperature was effectively uniform in all directions.

It was in fact the Pigeons that discovered CMB!

In 1964, radio astronomers Arno Penzias and Robert Wilson accidentally discovered CMB while mapping microwave signals received from the Milky Way. They thought the CMB signal was simply background noise, but it was consistent and would not go away no matter where the satellite would point to. They even cleaned pigeon droppings from the antennae equipment! They took their findings to Robert Dicke who had already theorised the existence of CMB. The group eventually concluded that what they received was a signal from a relic of the distant past.

NASA WMAP Mission 2001





Enhanced resolution of the fluctuations in CMB temperature. Showed that the CMB supported the Big Bang theory and our current understanding of the universe.

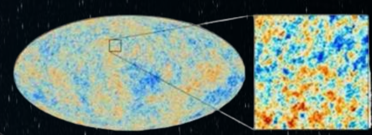
Glossary of Useful Terms

Anisotropy: A property which is non-uniform in different directions.

ΛCDM model: Lambda-Cold-Dark-Matter Model is what describes our current understanding of the universe.

Mean Free Path: The average distance a particle travels before experiencing a change as a result of collision.

ESA Planck Mission 2009

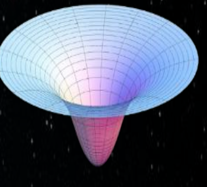


Produced the best resolution of CMB map and enhanced the visibility of temperature fluctuations, giving a deeper insight into the properties and origins of the universe.

Why are there fluctuations in CMB temperature?

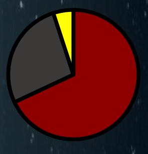
The temperature fluctuations arise due to uneven density distribution in the early universe. A denser region has a higher gravitational attraction. This can be shown as a deeper gravitational potential well. A photon trying to escape a deeper well would lose more energy, resulting in a cooler temperature in that region. These uneven regions of density became the precursors of galaxies and the large-scale structures we see in the universe today.

Visualisation of a typical gravitational potential well



What does the CMB tell us about the universe?

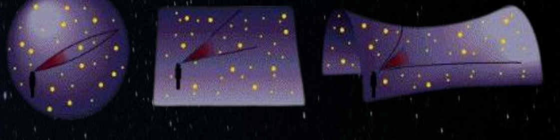
COMPOSITION



- Normal Matter 5%
- Dark Matter 27%
- Dark Energy 68%

CMB gives insight into the composition of the universe. The pie chart shows relative densities of the different constituents. This is known because the fluctuation pattern shown by CMB maps would change depending on how much of each constituent there is.

CURVATURE



CMB shows that the geometry of the universe is nearly flat. To learn how, check out: [WMAP- Shape of the universe (nasa.gov)]

ORIGIN

The largely uniform temperature of CMB indicates that all regions of the universe were close enough in the beginning to exchange energy, supporting the Big Bang model.

RATE OF EXPANSION

By observing the cosmic expansion rate at the time of CMB, the current expansion rate can be predicted based on how the universe has since cooled and evolved.